

The Constants of Nature Are Counts

A geometric hypothesis for the Standard Model parameters, and the date it can die

Brieuc de La Fournière | Arithmon program | arithmon.com

One compact seven-dimensional geometry. Zero adjustable parameters. Thirty-three exact algebraic relations among Standard Model constants, machine-checked in Lean 4 with no unproven step, from a ledger of inputs frozen and publicly deposited before the search began.

And one date. If DUNE measures the neutrino CP phase δ_{CP} outside $[182^\circ, 212^\circ]$ at 3σ , this framework is wrong, nothing in it can be tuned to survive, and the prediction on record is 197° . First beam is targeted for 2031.

The second paragraph is what earns the first. A framework with no free parameters cannot absorb a bad measurement. That is the cost of the claim, and it is the reason the claim is worth checking rather than dismissing.

The hypothesis. The dimensionless constants of the Standard Model are arithmetic and topological invariants of a compact manifold with G_2 holonomy, not free parameters to be fitted. The specific geometry has Betti numbers $(b_2, b_3) = (21, 77)$ and an $E_8 \times E_8$ motivated architecture. One relation shows what that means in a single line: the number of chiral fermion generations follows from the topological constraint $(\text{rank}(E_8) + N_{\text{gen}}) \times b_2 = N_{\text{gen}} \times b_3$, that is $(8 + N_{\text{gen}}) \times 21 = N_{\text{gen}} \times 77$, whose only integer solution is $N_{\text{gen}} = 3$. Not declared. Derived.

Continuously adjustable parameters	0 , after the input ledger is frozen
Parameter-free algebraic relations	33
Total derived observables	95 , of which 66 experimentally comparable
Agreement	11 exact to better than 0.01%, 53 within 1%
Formal verification	213 -conjunct Lean 4 certificate, 15 classified axioms, 0 unproven steps
Pre-registration	ledger, formula grammar and target list deposited before any search

What is not being claimed. Only the thirty-three parameter-free relations are direct consequences of the frozen ledger. The remaining observables are separated into declared categories: one-step extractions that use a named experimental anchor, multi-step chains conditional on the mechanism invoked, and structural diagnostics with no direct comparison. Counting all ninety-five as independent predictions would be the easiest available way to oversell this work, and the framework's own tables refuse to do it. Numerical precision is reported as a secondary figure: sub-percent agreement is exactly what a well-tuned fit also produces. The claim is structural, not numerical.

Why this is not numerology, as a commitment rather than an argument. The ledger, the formula grammar and the target list were frozen and deposited before any relation was searched for. The statistical procedure was fixed in advance too, and it is deliberately not bespoke: it is calibrated against historical verdicts, so that Eddington's fine-structure argument fails it and the quantum Hall relation passes it. A test only its author's framework can pass is not a test. Under that pre-registered null battery the honest outcome is modest and is reported as modest: one relation, m_H/m_W , survives at exact rank-one uniqueness, with Koide's $Q = 2/3$ close behind. Earlier versions led with joint coincidence probabilities around 10^{-346} ; those figures are retained as internal diagnostics and deliberately removed from the headline.

What would refute it.

Prediction	Value	Experiment	Timeline
δ_{CP}	197° , refuted outside $[182^\circ, 212^\circ]$ at 3σ	DUNE	beam targeted 2031
N_{gen}	3, refuted by a fourth generation	any	ongoing
m_s/m_d	20, current 20.0 ± 1.0	Lattice QCD	2028–2030
$\sin^2 \theta_W$	3/13	FCC-ee	2040s

What is open. Three questions are load-bearing and unresolved, stated at the same prominence as the results. Does the compact geometry exist? The global-analytic layer remains conditional on a declared hypothesis pack. Where does the gauge sector come from? The branching chain is compatible with Standard Model matter content but is not yet a physical breaking mechanism, and this is the largest gap in the work. Why $(21, 77)$? Five routes to deriving the Betti numbers were investigated and closed; one explicit residual remains, $b_2 + b_3 = 98 = \dim(K_7) \cdot \dim(G_2)$.

The question. Most proposals that attempt to explain the constants of Nature eventually fail. Perhaps this one will too. But the question does not go away with them: if the constants are not fundamental, they come from somewhere. This is one attempt to answer that while making every assumption explicit, every calculation reproducible, every prediction public, and every route to refutation available in advance. Whether the hypothesis survives is now a matter for mathematics and experiment, not for authority.

Full framework paper, formal sources, methodology and dated prediction scoreboard at arithmon.com and github.com/Arithmon. The underlying numerical work on G_2 metrics is cited as reference 25 in *Physics Letters B* **878** (2026) 140566. Every claim above is built to be checkable without reference to its author.